SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 2: 2. 8-Queen Problem

Aim: To place eight queens on a chessboard so that no two queens attack each other.

Python Program

```
N = 8

def solve(board, row):
    if row == N:
        return True
    for col in range(N):
        if all(board[r] != col and
               abs(board[r] - col) != row - r
               for r in range(row)):
            board[row] = col
            if solve(board, row + 1):
                return True
    return False

board = [-1] * N
if solve(board, 0):
    for r in range(N):
        print(" ".join("Q" if board[r] == c else "."
                       for c in range(N)))
else:
    print("No solution")
```

Sample Output

```
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```

Result: The program demonstrates the stated Artificial Intelligence problem.